ABDUL MOIZ

SP23-BSCS-0022

CM

ASSIGNMENT # 3

TRANSFORM AND CONQUER

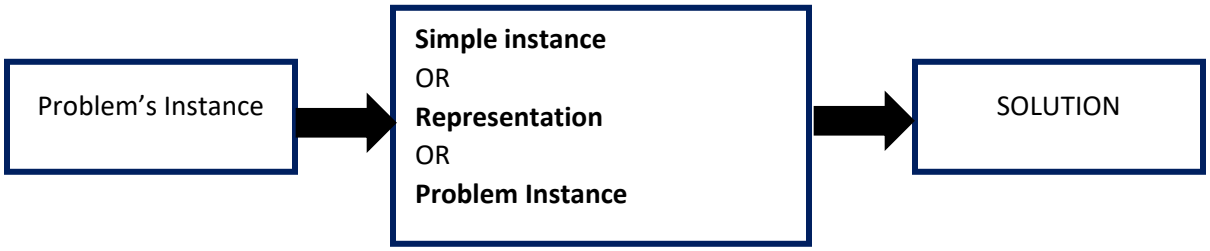
Transform and Conquer" is a method in algorithm design where we change a problem into another form that is easier to solve.

There are three ways to do this:

- ✓ **Instance Simplification:** Change the problem into an easier version of the same problem.
- ✓ **Representation Change:** Change the way the data is represented (without changing the problem itself).
- ✓ **Problem Reduction:** Convert the problem into a different one for which a solution already exists.

IN SIMPLE:

TRANSFORM AND CONQUER STRATEGY:



1. INSTANCE SIMPLIFICATION:

This means solving a problem more easily by simplifying the input first.

Example: Presorting:

Sorting a list first can make many tasks easier:

Searching for an item

Finding the median (middle value)

Checking if all elements are unique

Element Uniqueness Example:

Step 1: Sort the list (using Merge Sort or any efficient method)

Step 2: Go through the list and check if any two adjacent elements are the same

Time efficiency:

Using presorting: $\Theta(n \log n)$

Without sorting (brute force): $O(n^2)$

Computing Mode with Presorting:

Sort the list first.

Then count how often each number appears.

This is faster than comparing every number with every other one.

Searching with Presorting:

Sort the list first.

Then use binary search, which is much faster than linear search.

INSTANCE SIMPLIFICATION : PRESORTING

- Many questions can be answered if the lists are sorted.

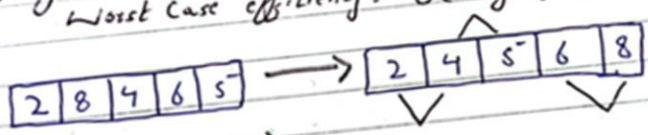
Time efficiency of the Problem's algorithm is dominated by the algorithm used for sorting the list.

EXAMPLE:

Checking element uniqueness in an array.

- Using brute force algorithm. Worst case efficiency $= O(n^2)$

- Using presorting : $n-1$ comparision.
  Worst case efficiency = $O(n \log n)$




```
Algo -> (A[0... n-1])
i = 0
Fmode = 0
while i ≤ n-1 do
    runlength = 1
    runValue = A[i]
        while i + runlength ≤ n-1
            and A[i+runlength] = runVal
    if runlength > Freq mode
        Fmode = runlength
    i = i + runlength
return
```

2. Representation Change

Changing the way the data is stored can help solve problems more efficiently.

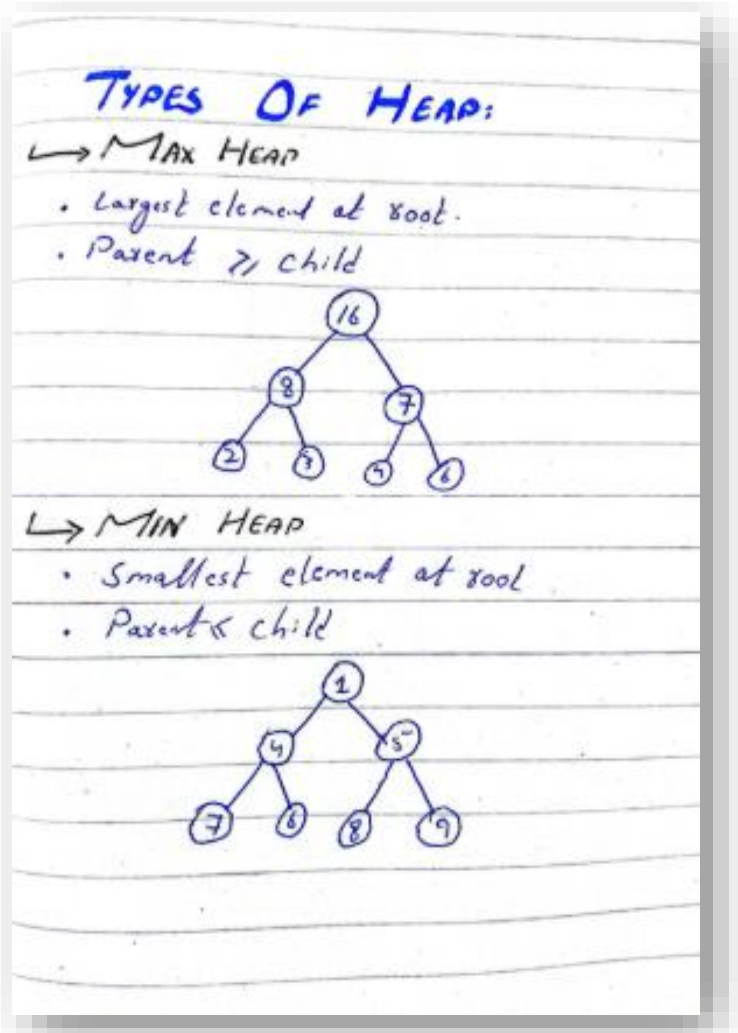
Example: Heap and Heapsort

A heap is a special kind of binary tree used to manage data efficiently.

Types of Heaps:

Max-Heap: Biggest value is at the top (root).

Min-Heap: Smallest value is at the top.



Heap Operations:

Insert: Add an item, then adjust the heap → O(log n)

Delete-Max: Remove the top item (max), fix the heap → O(log n)

Heapsort Steps:

Build a heap from the list.

Repeatedly remove the max item and adjust the heap.

Efficiency: Θ(n log n)

In-place sort (doesn't need extra space)

Not stable (equal elements might change order)

Horner's Rule for Polynomial Evaluation

Problem:

Evaluate polynomial:

$p(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots + a_1 x + a_0$

Brute-force Methods:

Use nested loops or repeated multiplications , Complexity: $O(n^2)$.

Priority Queue Implementation Using Heaps

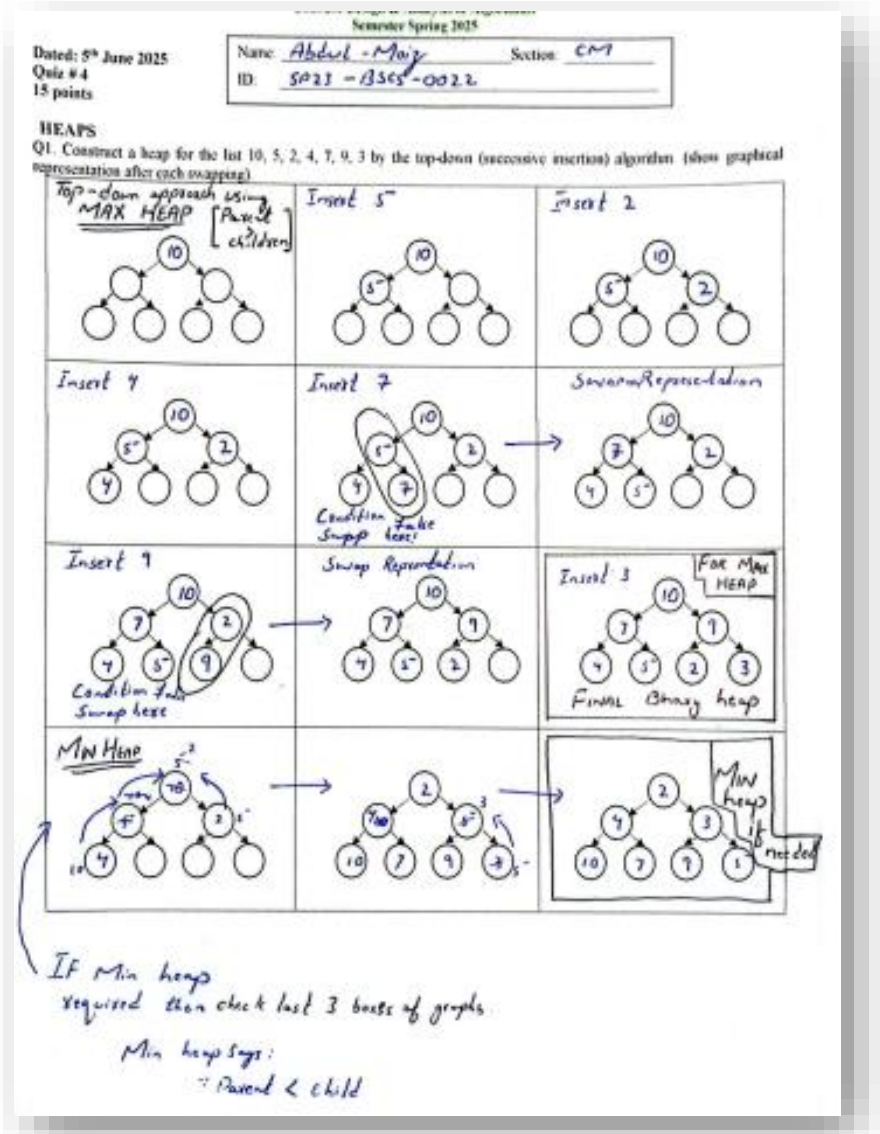
Definition:

A Priority Queue is an Abstract Data Type (ADT) where each element has a priority.

Comparison of Implementations:

Operation	Unsorted Array	Sorted List	Heap
Insert	O(1)	O(n)	O(log n)
FindMin	O(n)	O(1)	O(1)
RemoveMin	O(n)	O(1)	O(log n)

HERE THE PERFECT EXAMPLE OF THE INSERTION IN HEAP BY USING BOTH TYPES MAX AND MIN HEAP FROM OUR QUIZ:



Semester Spring 2025

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Quiz # 4
15 points

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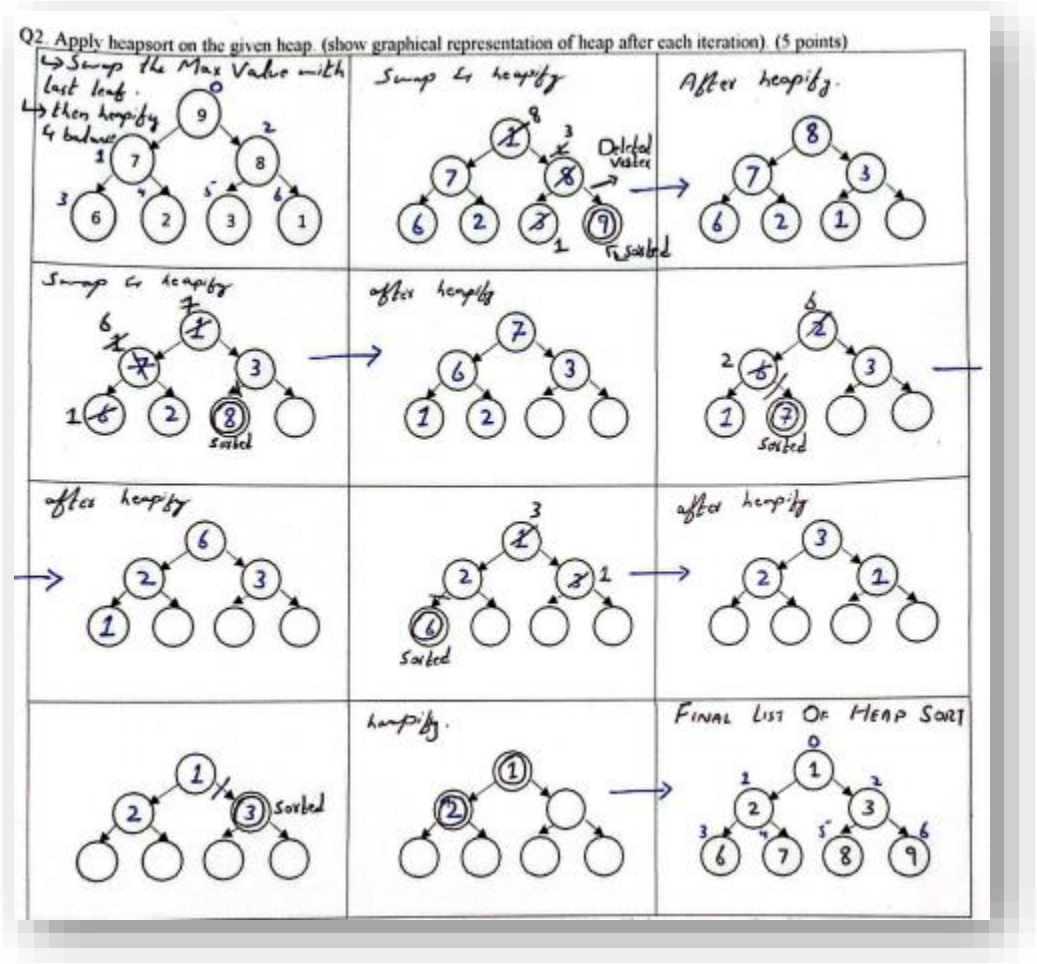
HEAPS

Q1. Construct a heap for the list 10, 5, 2, 4, 7, 9, 3 by the top-down (successive insertion) algorithm (show graphical representation after each swapping)

HERE THE PERFECT EXAMPLE OF THE HEAP SORT WITH SIMULATION
FROM OUR QUIZ:



Horner's Rule for Polynomial Evaluation

Problem:

Evaluate polynomial:

$p(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots + a_1 x + a_0$

Example : Horner's Rule

- Horner's rule gives a method to evaluate polynomials.

$\sum$ ai x^i

```
public static int horner (int[], int n, int x)
{
    int result = a [n];
    for (int j = n-1; j > = 0; j--)
        result = result * c + a[j];
    return result;
}
```

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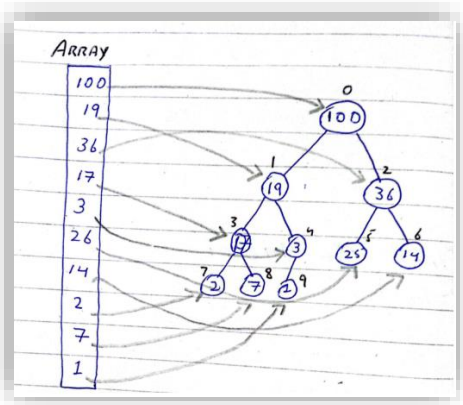


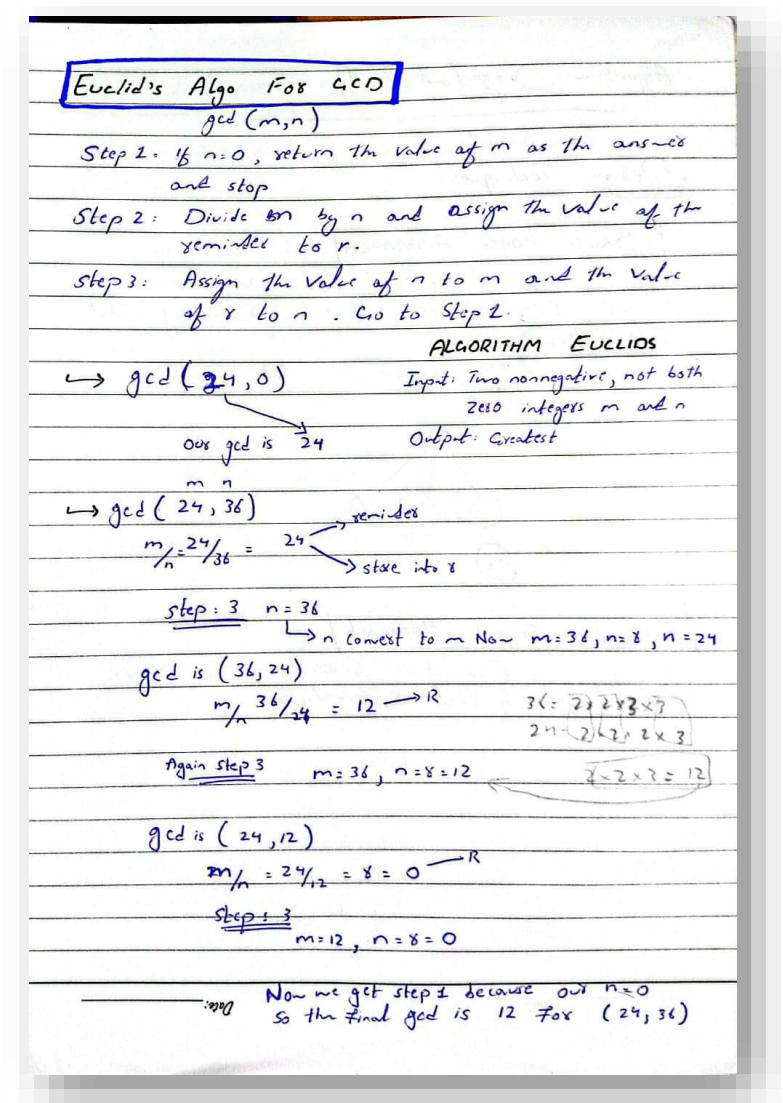
3. Problem Reduction

Turn a problem into another different problem that we already know how to solve.

GCD Using Euclid's Algorithm (Problem Reduction Example)

When we want to find the Greatest Common Divisor (GCD) of two numbers a and b, instead of checking all possible divisors, we can reduce the problem using a simple rule:

GCD(a, b) = GCD(b, a mod b):



Euclid's Algo For GCD

gcd (m,n)

Step 1: If n=0, return the value of m as the answer and stop

Step 2: Divide m by n and assign the value of the reminder to r.

Step 3: Assign the value of n to m and the value of r to n. Go to Step 2.

ALGORITHM EUCLIDS

Input: Two nonnegative, not both zero integers m and n

Output: Greatest

→ gcd (24, 0)

our gcd is 24

m n

→ gcd (24, 36)

m/n = 24/36 = 24 → reminder, → store into r

step: 3 n = 36

→ n convert to m Now m=36, n=r, n=24

gcd is (36, 24)

m/n 36/24 = 12 → R

36 = 2, 2×3×3

24 = 2, 2×2×3

Again step 3 m=36, n=r=12

r = 2×3 = 12

gcd is (24, 12)

m/n = 24/12 = r = 0 → R

Step: 3

m=12, n=r=0

Now we get step 1 because our n=0 so the final gcd is 12 for (24, 36)